

19-205-0604:Heat and Mass Transfer

Dr. James Varghese School of Engineering, CUSAT

Module I

Introduction to heat transfer – basic modes of heat transfer – conduction heat transfer –Fourier law of heat conduction– temperature dependence of thermal conductivity- general heat conduction equation in cartesian, cylindrical and spherical coordinates – boundary conditions – one-dimensional steady state conduction- critical insulation thickness -one-dimensional steady state conduction with heat generation -extended surface – two dimensional steady state heat conduction – conduction shape factor – unsteady state heat conduction in one dimension – lumped heat capacity system – semi-infinite solids with sudden change in surface temperature – Introduction to numerical methods in conduction.

Module II

Convective heat transfer – Newton's law of cooling – Prandtl number – laminar forced convection heat transfer from flat plates – fully developed laminar flow in pipes – turbulent forced convection – Reynolds' analogy – natural convection – natural convection heat transfer from vertical plates and horizontal tubes – condensation and boiling – film and drop wise condensation – film boiling and pool boiling – introduction to multiphase flow and heat transfer. Diffusion and convective mass transfer-Ficks law of diffusion.

Module III

Radiative transfer – electromagnetic radiation spectrum – thermal radiation –radiation properties- black body, gray body – monochromatic and total emissive power – Planck's law – Stefan-Boltzman law – Wien's displacement law – Kirchhoffs identity – shape factor- reciprocity relation – heat exchange between non black bodies; surface and shape resistances- electrical network analogy- heat transfer between parallel surfaces – radiation shields.

Module IV

Heat Exchangers: Type of heat exchangers- overall heat transfer coefficient -fouling factors -Logarithmic mean temperature difference (LMTD)- derivation of LMTD for parallel flow and counter flow heat exchangers-LMTD correction factor- effectiveness, NTU method of heat exchanger analysis- effectiveness derivation for parallel flow and counter flow heat exchangers. Design of parallel flow-counterflow-shell and tube multipass heat exchangers-condensers.

References:

1. Cengel, Heat Transfer, 3rd edition, Tata Mc Graw Hill, (2007).
2. Holman J. P., Heat Transfer, 10th edition, McGraw Hill International Students Edition, (2009).
3. Incropera F. P. & De Witt, D. P., Fundamentals of Heat and Mass Transfer, 7th Ed., Wiley, (2011).
4. Kreith F., Heat Transfer, International Text Book Company, (1958).
5. Gebhart B., Heat Transfer, 2nd edition, McGraw Hill, (1971).
6. Rajput, R. K., Heat and Mass Transfer, S Chand, (2007).
7. Venkanna, Fundamentals of HMT, Prentice Hall, (2011).

Data Book:

1. C. P. Kothandaraman, Heat & Mass Transfer Data Book, 8th edition, New Age International, (2014).
2. Domkundwar, Heat & Mass Transfer Data Book, 3rd edition, Dhanpat Rai, (2006).

Approved data book is to be specified in the question paper.

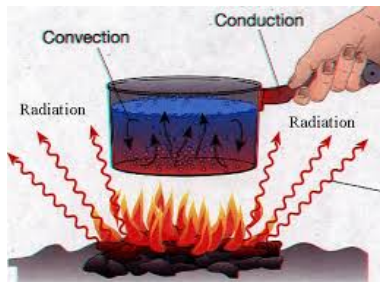
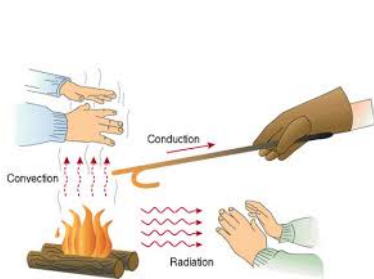
Course Outcome

1. Understand the basic modes of heat transfer
2. Get an insight on conduction, convection and radiation heat transfers
3. Apply the knowledge of heat transfer through various geometries for optimizing the thickness of insulation.
4. Understand the unsteady heat conduction and its applications.
5. Get a concept on multiphase flow, diffusion and convective mass transfer
6. Attain information of parallel and counterflow heat exchangers and their design aspects

Introduction-Heat Transfer

- ① Heat $\implies$ Energy Transport rate (kW or kJ/s)
- ② Thermodynamics $\implies$ Relation between heat and other forms of energy
- ③ Heat transfer rate form of Energy transfer
- ④ Modes of heat transfer
 - ① Conduction
 - ② Convection
 - ③ Radiation
- ⑤ Potential $\implies$ Temperature difference (K)

Conduction heat transfer



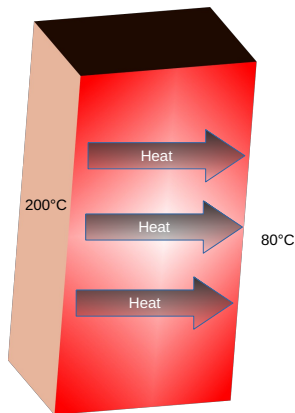
Conduction heat transfer

- 1 Heat transfer takes place as conduction in a solid if there is a temperature gradient.
- 2 conduction is mediated by the combination of vibrations and collisions of molecules
- 3 diffusion and collisions of free electrons
- 4 Conduction is within a solid or between solid objects in thermal contact.

Conduction heat transfer

Fourier law of Conduction

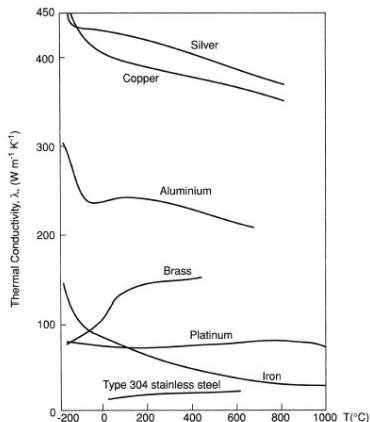
- 1 Conduction heat transfer Q (W)
- 2 $Q = -kA \frac{dT}{dx}$
- 3 k - Thermal conductivity (W/mK)
- 4 A cross sectional area through which heat transfer occurs
- 5 -ve sign heat flow direction



Thermal conductivity

Thermal Conductivity of Common Materials (at 25°C)

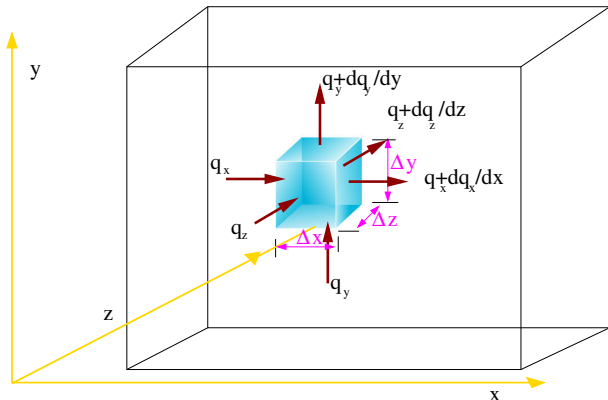
Material	Conductivity (Watts/meter-°C)
Acrylic	0.200
Air	0.024
Aluminum	250.000
Copper	401.000
Carbon Steel	54.000
Concrete	1.050
Glass	1.050
Gold	310.000
Nickel	91.000
Paper	0.050
PTFE (Teflon [®])	0.250
PVC	0.190
Silver	429.000
Steel	46.000
Water	0.580



Conduction heat transfer

- 1 Less thermal conductivity materials -insulators
- 2 Conductivity changes with temperature $k(T) = k_0(1 + \beta T)$

General Heat Conduction equation-Rectangular coordinates



- 1 Elemental volume dx, dy, dz
- 2 Thermal conductivity k , density ρ , specific heat capacity c

General Heat Conduction equation-Rectangular coordinates

- 1 $T = T(x, y, z, \tau)$
- 2 Energy entering in x direction $Q_x = -kA \frac{\partial T}{\partial x}$
- 3 Energy leaving in x direction $Q_{x+\Delta x} = Q_x + \frac{dQ_x}{dx} \Delta x$
- 4 Energy leaving in x direction $= -kA \frac{\partial T}{\partial x} + -kA \frac{\partial^2 T}{\partial x^2} \Delta x$
- 5 Heat generation per unit volume $= \dot{q} \text{ (W/m}^3\text{)}$
- 6 Total heat generation $Q_{gen} = \dot{q} \Delta x \Delta y \Delta z$
- 7 Mass of the elemental volume $= \rho \Delta x \Delta y \Delta z$
- 8 Internal energy change of the element $I.E., (mc \Delta T) = \rho c \Delta x \Delta y \Delta z \frac{\partial T}{\partial \tau}$
- 9 energy balance
$$Q_x + Q_y + Q_z + Q_{gen} = I.E. + Q_{x+\Delta x} + Q_{y+\Delta y} + Q_{z+\Delta z}$$
- 10 General heat conduction equation $\frac{\partial^2 T}{\partial x^2} + \frac{\partial^2 T}{\partial y^2} + \frac{\partial^2 T}{\partial z^2} + \frac{\dot{q}}{k} = \frac{1}{\alpha} \frac{\partial T}{\partial \tau}$

General Heat Conduction equation-Rectangular coordinates

- Simplifications

- 1 General heat conduction equation $\frac{\partial^2 T}{\partial x^2} + \frac{\partial^2 T}{\partial y^2} + \frac{\partial^2 T}{\partial z^2} + \frac{\dot{q}}{k} = \frac{1}{\alpha} \frac{\partial T}{\partial \tau}$
- 2 With no heat generation $\frac{\partial^2 T}{\partial x^2} + \frac{\partial^2 T}{\partial y^2} + \frac{\partial^2 T}{\partial z^2} = \frac{1}{\alpha} \frac{\partial T}{\partial \tau}$
- 3 Steady state conduction $\frac{\partial^2 T}{\partial x^2} + \frac{\partial^2 T}{\partial y^2} + \frac{\partial^2 T}{\partial z^2} = 0$
- 4 Two dimensional steady state conduction $\frac{\partial^2 T}{\partial x^2} + \frac{\partial^2 T}{\partial y^2} = 0$
- 5 One dimensional steady state conduction $\frac{\partial^2 T}{\partial x^2} = 0$
- 6 Assumes conductivity and density constant.
- 7 α is the thermal diffusivity

General Heat Conduction equation-Cylindrical coordinates

1 Cylindrical coordinates

$$T = T(r, \theta, z)$$

2

$$\frac{\partial^2 T}{\partial r^2} + \frac{1}{r} \frac{\partial T}{\partial r} + \frac{1}{r^2} \frac{\partial^2 T}{\partial \theta^2} + \frac{\partial^2 T}{\partial z^2} + \frac{\dot{q}}{k} = \frac{1}{\alpha} \frac{\partial T}{\partial \tau}$$

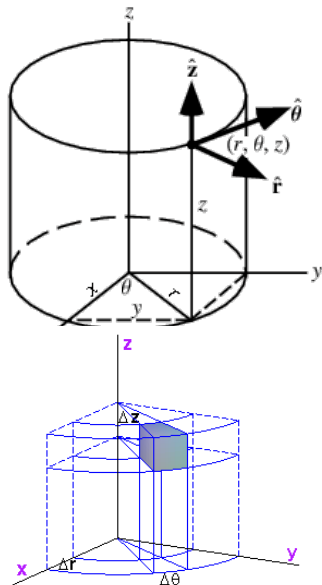
3 One dimensional steady state

$$\frac{d^2 T}{dr^2} + \frac{1}{r} \frac{dT}{dr} = 0$$

4 Heat transfer in radial direction

only $Q = -2\pi krL \frac{dT}{dr}$

5 Useful to handle radial systems



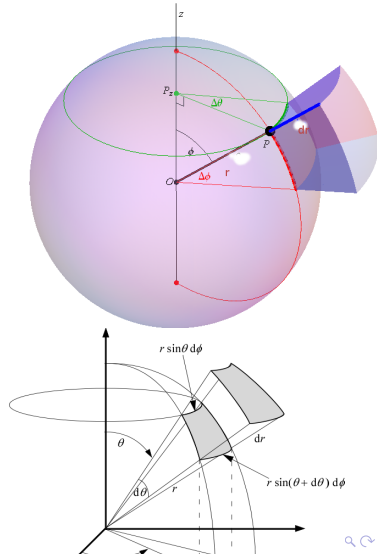
General Heat Conduction equation-Spherical coordinates

1 Spherical coordinates

$$T = T(r, \theta, \phi)$$

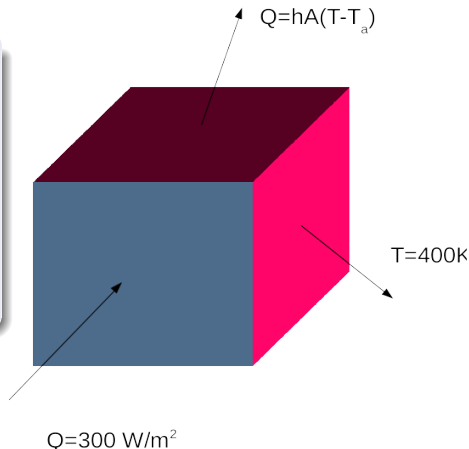
2

$$\frac{1}{r} \frac{\partial^2(rT)}{\partial r^2} + \frac{1}{r^2 \sin \theta} \frac{\partial}{\partial \theta} \left(\sin \theta \frac{\partial T}{\partial \theta} \right) + \frac{1}{r^2 \sin^2 \theta} \frac{\partial^2 T}{\partial \phi^2} + \frac{\dot{q}}{k} = \frac{1}{\alpha} \frac{\partial T}{\partial \tau}$$



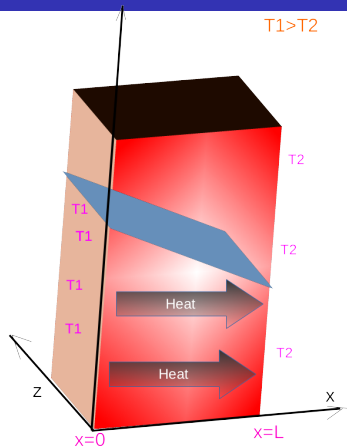
Boundary conditions

- 1 Constant temperature $T = T_1...$
- 2 Constant heat flux $q = Q/A$ (W/m^2)
- 3 Convective heat transfer
 $Q = hA_s(T - T_\infty)$
- 4 Radiative heat transfer
- 5 Combined



One dimensional steady heat transfer

- 1 One dimensional temperature variation
 $T = T(x)$
- 2 $\frac{d^2 T}{dx^2} = 0$
- 3 Integrating $\frac{dT}{dx} = C_1$
- 4 Integrating $T = C_1 x + C_2$
- 5 C_1, C_2 depends on the boundary conditions.
- 6 For b.c at $x = 0, T = T_1$ and $x = L, T = T_2$
- 7 $T = \frac{(T_2 - T_1)x}{L} + T_1$
- 8 Heat transfer $Q = \frac{kA(T_1 - T_2)}{L}$



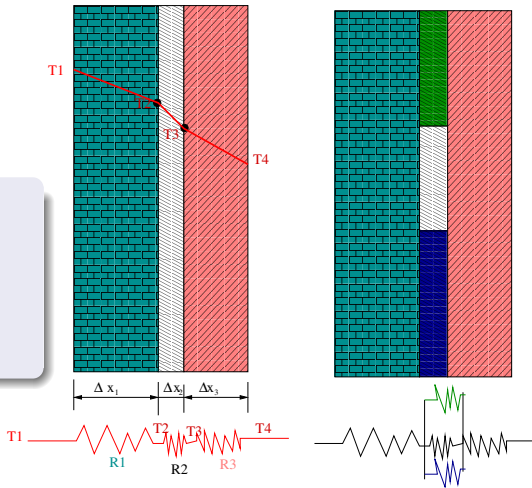
Composite walls

1 Plates in series - parallel

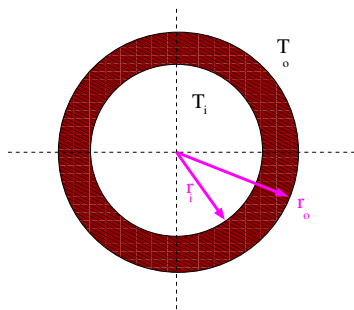
2 Same heat flow

$$3 \quad Q = \frac{T_1 - T_4}{\Delta x_1 / k_1 A + \Delta x_2 / k_2 A + \Delta x_3 / k_3 A}$$

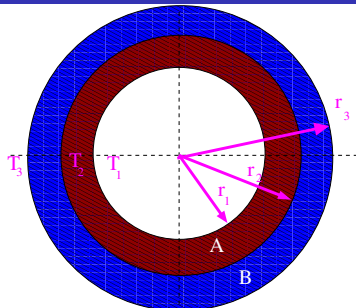
$$4 \quad Q = \frac{\Delta T_{overall}}{\Sigma R_{overall}}$$



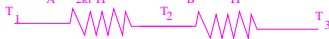
One dimensional steady heat transfer -Cylinders



$$R = \frac{\ln(r_o/r_i)}{2kl\Pi}$$

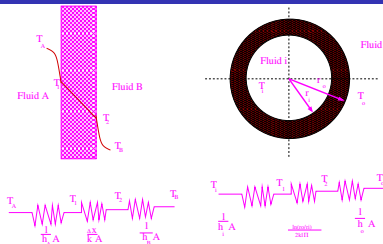


$$R_A = \frac{\ln(r_2/r_i)}{2kl\Pi} \quad R_B = \frac{\ln(r_3/r_2)}{2kl\Pi}$$



- ① One dimensional temperature variation $T = T(r)$
- ② Radial heat transfer $Q_r = \frac{2\pi kL(T_i - T_o)}{\ln(r_o/r_i)}$
- ③ Composite cylinder $Q_r = \frac{2\pi L(T_1 - T_3)}{\ln(r_2/r_1)/k_A + \ln(r_3/r_2)/k_B}$

Overall heat transfer -coefficient



① Heat transfer $Q = \frac{T_A - T_B}{1/h_A A + \Delta x/k A + 1/h_B A}$, $Q = U A \Delta T_{overall}$

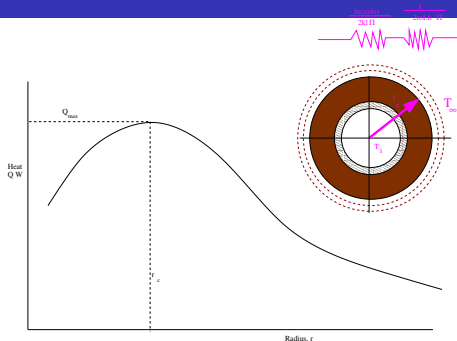
② Overall heat transfer coefficient $U = \frac{1}{1/h_A + \Delta x/k + 1/h_B}$

③ Cylinder $Q = \left[\frac{T_i - T_o}{\frac{1}{h_i A_i} + \frac{\ln(r_o/r_i)}{2\pi k L} + \frac{1}{h_o A_o}} \right]$

④ $Q = U_o A_o \Delta T_{overall}$, $U_o = \frac{1}{\frac{A_o}{A_i} \frac{1}{h_i} + \frac{A_o \ln(r_o/r_i)}{2\pi k L} + \frac{1}{h_o}}$

⑤ $Q = U_i A_i \Delta T_{overall}$, $U_i = \frac{1}{\frac{1}{h_i} + \frac{A_i \ln(r_o/r_i)}{2\pi k L} + \frac{A_i}{A_o} \frac{1}{h_o}}$

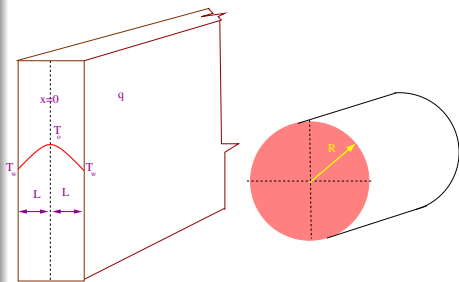
Critical radius of insulation



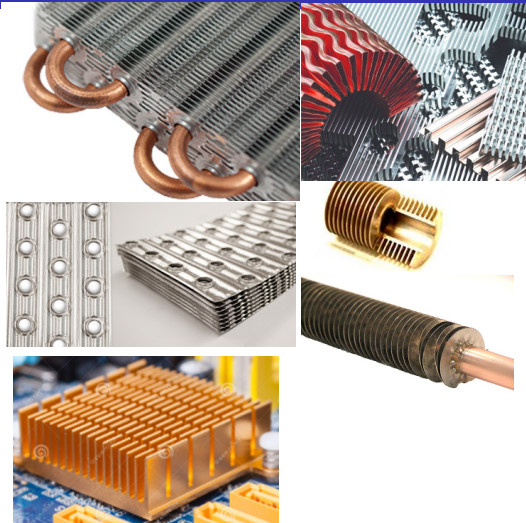
- 1 Heat transfer $Q = \frac{2\pi L(T_i - T_\infty)}{\frac{\ln(r_o/r_i)}{k} + \frac{1}{r_o h}}$
- 2 $\frac{dq}{dr_o} = 0 = \frac{-2\pi L(T_i - T_\infty) \left(\frac{1}{kr_o} - \frac{1}{hr_o^2} \right)}{\left[\frac{\ln(r_o/r_i)}{k} + \frac{1}{r_o h} \right]^2}$
- 3 Critical radius of insulation $r_c = \frac{k}{h}$

Plane wall with heat source

- 1 $\frac{d^2 T}{dx^2} + \frac{q}{k} = 0$
- 2 Boundary conditions $T = T_w$,
at $x = \pm L$, $T = T_o$ at $x = 0$
- 3 Solution $\frac{T - T_w}{T_o - T_w} = 1 - \left(\frac{x}{L}\right)^2$
- 4 Alternatively $\frac{T - T_o}{T_w - T_o} = \left(\frac{x}{L}\right)^2$
- 5 Cylinder with heat generation
 $T = T_w$, at $-r = R$
- 6 Solution $T - T_w = \frac{q}{4k}(R^2 - r^2)$
- 7 Alternatively $\frac{T - T_w}{T_o - T_w} = 1 - \left(\frac{r}{R}\right)^2$
- 8 $T_o = \frac{qR^2}{4k} + T_w$



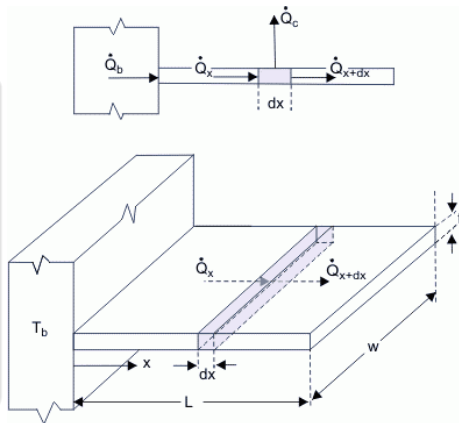
Extended surfaces -Fins



- 1 Need for Extended surfaces
- 2 Applications

Heat transfer from fins

- 1 Base temperature T_b
- 2 Ambient temperature T_∞
- 3 Convection heat transfer coefficient $h(W/m^2K)$
- 4 Convective heat transfer $Q = hA_s(T - T_\infty)$
- 5 Surface perimeter P
- 6 Surface area $A_s = Pdx$



Heat transfer from fins

① $Q_x = -kA \frac{dT}{dx}$

② $Q_{x+dx} = Q_x + -kA \frac{d^2T}{dx^2} dx$

③ Energy balance

$$Q_x = Q_{x+dx} + hPdx(T - T_\infty)$$

④ $\Theta = T - T_\infty$

⑤ $kA \frac{d^2\Theta}{dx^2} - hP\Theta = 0$

⑥ Fin solution

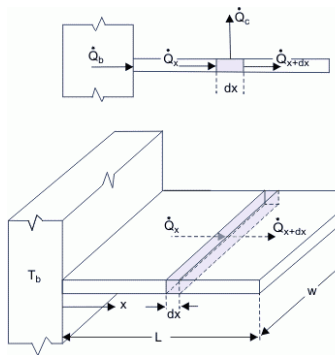
$$\Theta = T - T_\infty = C_1 e^{-mx} + C_2 e^{mx}$$

⑦ Fin very long

$$\Theta = \Theta_0 - at - x = 0, \Theta = 0 - at - x = \infty$$

⑧ $\frac{T - T_\infty}{T_0 - T_\infty} = e^{-mx}, m = \sqrt{\frac{hP}{kA}}$

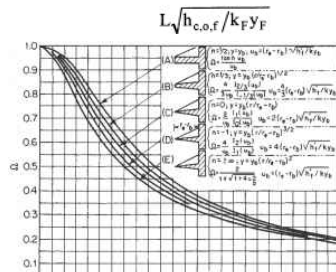
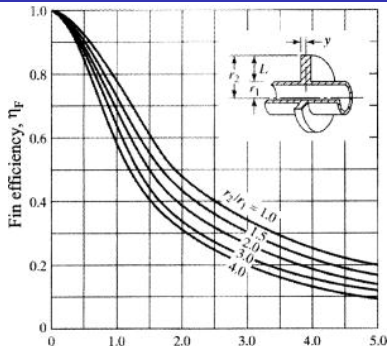
⑨ Heat loss $q = -kA \frac{\partial T}{\partial x} \Big|_{x=0} = \Theta_0 \sqrt{hPkA}$



Fin Efficiency and effectiveness

- 1 Fin efficiency $-\eta_{fin} = \frac{Q - \text{Fin-actual}}{Q - \text{Fin-entire-fin-at-base-temp.}}$
- 2 Fin conduction resistance Vs convective resistance
- 3 Fin thickness varying - minimise weight
- 4 Fin effectiveness -

$$\varepsilon_{fin} = \frac{Q - \text{Fin-actual}}{Q - \text{with-out-fin-(base-area)}}$$
- 5 Fin effectiveness must be > 1



Steady state two dimensional

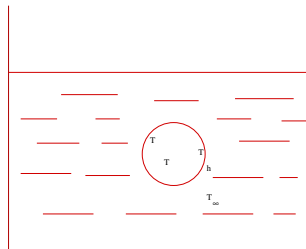
Two dimensional

- 1 $\frac{\partial^2(T)}{\partial x^2} + \frac{\partial^2(T)}{\partial y^2} = 0$
- 2 Heat flow $q_x = -kA_x \frac{\partial T}{\partial x}$
- 3 Heat flow $q_y = -kA_y \frac{\partial T}{\partial y}$
- 4 Two dimensional system with two temperatures $q = kS\Delta T_{overall}$
- 5 S- conduction shape factor

Lumped System Analysis

Assumptions

- 1 Interior temperature is uniform
- 2 One dimensional
- 3 h - constant
- 4 k - constant
- 5 h is low compared to k

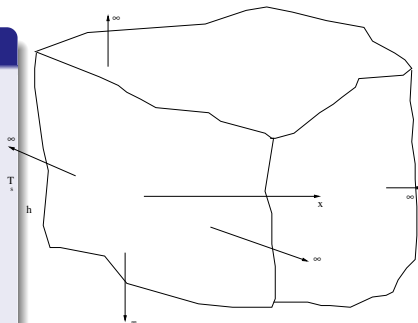


- ① Heat transfer by convection = Internal energy change
- ② $hA(T - T_{\infty}) = -mc_p dT/dt$
- ③ $\frac{d(T - T_{\infty})}{(T - T_{\infty})} = -\frac{hA_s dt}{\rho V c_p} = \frac{d\theta}{\theta} = -\frac{hA_s dt}{\rho V c_p}$
- ④ Limits $\theta = \theta_i \Rightarrow (T_i - T_{\infty}), t = 0$ and $\theta = \theta_t \Rightarrow (T - T_{\infty}), t = t$
- ⑤ integrating $\int_{\theta_i}^{\theta_t} \frac{d(\theta)}{\theta} = -\int_0^t \frac{hA_s dt}{\rho V c_p}$
- ⑥ $[ln(\theta)]_{\theta_i}^{\theta_t} = ln[\theta_t] - ln[\theta_i] = -\frac{hA_s t}{\rho V c_p}$
- ⑦ $ln\left[\frac{\theta_t}{\theta_i}\right] = -\frac{hA_s t}{\rho V c_p}$
- ⑧ $\frac{(T - T_{\infty})}{(T_i - T_{\infty})} = e^{-bt}$, where $b = \frac{hA_s}{\rho V c_p}$
- ⑨ Criterion Biot Number $Bi = hL/k \leq 0.1$, Error within 5%
- ⑩ $\frac{(T - T_{\infty})}{(T_i - T_{\infty})} = e^{-BiFo}$, $Bi = \frac{hL}{k}$, $Fo = \frac{\alpha t}{L^2}$

Unsteady conduction semi infinite solids

Semi infinite solid

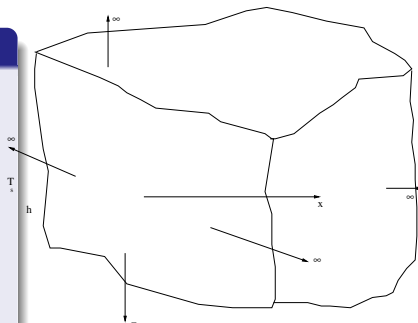
- 1 $\frac{\partial^2 T}{\partial x^2} = \frac{1}{\alpha} \frac{\partial T}{\partial t}$
- 2 $T(0, t) = T_s, T(x \rightarrow \infty, t) = T_i$
- 3 initial condition $T(x, 0) = T_i$
- 4 Use similarity variable $\eta = \frac{x}{\sqrt{4\alpha t}}$
- 5 boundary condition $\eta = 0$, at $x = 0, \eta \rightarrow \infty$ at $x \rightarrow \infty$
- 6 after simplification $\frac{d^2 T}{d\eta^2} = -2\eta \frac{dT}{d\eta}$
- 7 $T(0) = T_s$ and $T(\eta \rightarrow \infty) = T_i$



Unsteady conduction semi infinite solids

Semi infinite solid contd

- 1 define a new variable $w = \frac{dT}{d\eta}$
- 2 $\frac{dw}{d\eta} = -2\eta w$
- 3 $\ln w = -\eta^2 + C_0$
- 4 $w = C_1 e^{-\eta^2}$, where, $C_1 = \ln C_0$
- 5 substituting $w = \frac{dT}{d\eta}$ and integrating
- 6 $T = C_1 \int_0^{\eta} e^{-u^2} du + C_2$
- 7 where 'u' is a dummy variable The boundary condition $\eta = 0$ gives $C_2 = T_s$



semi infinite solids- unsteady conduction

- ① applying boundary condition $\eta \longrightarrow \infty$
- ② $T_i = C_1 \int_0^{\infty} e^{-u^2} du + C_2 = C_1 \frac{\sqrt{\pi}}{2} + T_s$
- ③ give $C_1 = \frac{2(T_i - T_s)}{\sqrt{\pi}}$
- ④ substituting C_1 and C_2
- ⑤ $\frac{T - T_s}{T_i - T_s} = \frac{2}{\sqrt{\pi}} \int_0^{\eta} e^{-u^2} du = \text{erf}(\eta) = 1 - \text{erfc}(\eta)$
- ⑥ where $\text{erf}(\eta) = \frac{2}{\sqrt{\pi}} \int_0^{\eta} e^{-u^2} du$ and $\text{erfc}(\eta) = 1 - \frac{2}{\sqrt{\pi}} \int_0^{\eta} e^{-u^2} du$
- ⑦ heat flux at the surface $q_s = -k \frac{\partial T}{\partial x} \big|_{x=0} = -k \frac{dT}{d\eta} \frac{\partial \eta}{\partial x} \big|_{x=0}$
- ⑧ $= -k C_1 e^{-\eta^2} \frac{1}{\sqrt{4\alpha t}} \big|_{\eta=0} = \frac{k(T_s - T_i)}{\sqrt{\pi\alpha t}}$

semi infinite solids- unsteady conduction

① specified surface temperature $T_s = C$

②
$$\frac{T(x,t)-T_i}{T_s-T_i} = \operatorname{erfc}\left(\frac{x}{2\sqrt{\alpha t}}\right)$$

③
$$q(t) = \frac{k(T_s-T_i)}{\sqrt{\pi\alpha t}}$$

④ Specified surface heat flux $q_s = C$

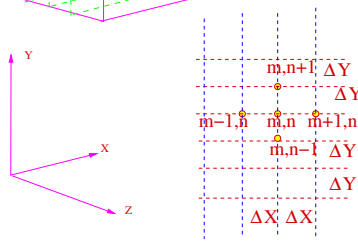
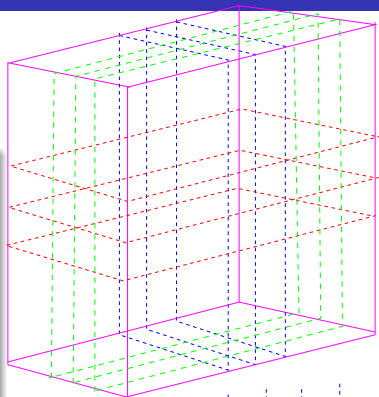
⑤
$$T(x,t) - T_i = \frac{q_s}{k} \left[\sqrt{\frac{4\alpha t}{\pi}} \exp\left(\frac{-x^2}{4\alpha t}\right) - x \operatorname{erfc}\left(\frac{x}{2\sqrt{\alpha t}}\right) \right]$$

⑥ convection on the surface $q_s(t) = h(T_\infty - T(0,t))$

⑦
$$\frac{T(x,t)-T_i}{T_\infty-T_i} = \operatorname{erfc}\left(\frac{x}{2\sqrt{\alpha t}}\right) - \exp\left(\frac{hx}{k} + \frac{h^2\alpha t}{k^2}\right) \operatorname{erfc}\left(\frac{x}{2\sqrt{\alpha t}} + \frac{h\sqrt{\alpha t}}{k}\right)$$

Numerical methods in conduction

- 1 Problem solutions $\Rightarrow$ Analytical, Experimental, Numerical
- 2 Numerical $\Rightarrow$ FDM, FEM, Finite Volume
- 3 FDM - Divide domain into nodes
- 4 Governing equations (PDE's) converted to algebraic equations
- 5 Write equations for each nodes



Numerical methods in conduction

- ① At the node $(m+1/2, n)$ $\frac{\partial T}{\partial x} = \frac{T_{m+1,n} - T_{m,n}}{\Delta x}$
- ② At the node $(m-1/2, n)$ $\frac{\partial T}{\partial x} = \frac{T_{m,n} - T_{m-1,n}}{\Delta x}$
- ③ At the node $(m, n+1/2)$ $\frac{\partial T}{\partial y} = \frac{T_{m,n+1} - T_{m,n}}{\Delta y}$
- ④ At the node $(m, n-1/2)$ $\frac{\partial T}{\partial y} = \frac{T_{m,n} - T_{m,n-1}}{\Delta y}$
- ⑤ At the node (m, n) $\frac{\partial^2 T}{\partial x^2} = \left[\frac{\left[\frac{\partial T}{\partial x} \right]_{m+1/2,n} - \left[\frac{\partial T}{\partial x} \right]_{m-1/2,n}}{\Delta x} \right]$
- ⑥ At the node (m, n) $\frac{\partial^2 T}{\partial x^2} = \frac{T_{m+1,n} + T_{m-1,n} - 2T_{m,n}}{\Delta x^2}$
- ⑦ At the node (m, n) $\frac{\partial^2 T}{\partial y^2} = \left[\frac{\left[\frac{\partial T}{\partial y} \right]_{m,n+1/2} - \left[\frac{\partial T}{\partial y} \right]_{m,n-1/2}}{\Delta y} \right]$
- ⑧ At the node (m, n) $\frac{\partial^2 T}{\partial y^2} = \frac{T_{m,n+1} + T_{m,n-1} - 2T_{m,n}}{\Delta y^2}$
- ⑨ 2D equation
$$\frac{\partial^2 T}{\partial x^2} + \frac{\partial^2 T}{\partial y^2} = \frac{T_{m+1,n} + T_{m-1,n} - 2T_{m,n}}{\Delta x^2} + \frac{T_{m,n+1} + T_{m,n-1} - 2T_{m,n}}{\Delta y^2} = 0$$
- ⑩ Solution - Gauss - Seidel Iteration etc.